

Artificial Intelligence (AI2002)

Final Exam

Date: 26/05/2025

Total Time: 3 hr

Course Instructors

Total Marks: 50

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Total Questions: 5

Roll No

Section

Student Signature

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ATTEMPT ALL QUESTIONS AND THEIR PARTS IN SEQUENCE

CLO1: Summarize the notions of rational behavior and intelligent agents.

Q1. Give short answers to the following in two lines:

[Marks 10, 30 min]

A. What are the fundamental components essential for an intelligent agent to function?

Perception, Learning Reasoning, Action, knowledge Representation -Any two for M1

B. What is an actuator in an intelligent agent?

A device/mechanism that allows an agent to perform actions (physical/software).

C. What is Rationality in the context of an intelligent agent?

Acts to maximize performance measures based on (Prior knowledge and Percept history/Under uncertainty, maximizes expected utility (e.g., probabilistic outcomes).

D. What is the primary role of an AI problem-solving agent?

To find a sequence of actions to reach a goal state/optimal sequence of actions.

E. Why is careful problem formulation crucial before applying search algorithms?

Precisely defines initial state, actions and goal/impacts search efficiency/impacts search effectiveness. -Any one for M1

F. Consider a taxi-driving scenario, where the traffic conditions and actions of other drivers are unpredictable. List all the environment types this scenario belongs to with reasons.

Stochastic (unpredictable traffic/drivers), Partially Observable (incomplete info about others/traffic), Multi-agent (other drivers are agents), Sequential (current actions affect future states), Dynamic (environment changes during deliberation), Continuous (time, speed, location values). -Any four for M1

G. How does a heuristic function improve a search algorithm compared to uninformed search?

Provides an estimate of cost/distance to goal/Prioritizes more promising paths/Explores search space more efficiently. -Anyone for M1

H. What is the difference between an admissible and inadmissible heuristic?

Admissible: Never overestimates true cost i.e. Admissible: $\leq$ actual cost.

Inadmissible: Overestimates true cost/can lead to non-optimal solutions.

I. Discuss how alpha-beta pruning improves the efficiency of Minimax Search algorithm?

Uses alpha (α) and beta (β) bounds to Prune branches where MIN/MAX cannot improve the current decision/Preserve optimality while evaluating fewer nodes/Cuts off branches where MIN's best move is worse than MAX's current α . -Any one for M1

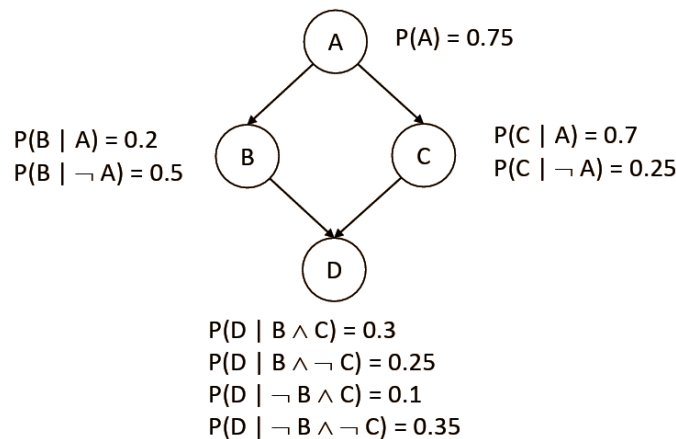
J. Explain the role of mutation and crossover in an evolutionary algorithm.

Mutation: Introduces random changes in solutions/Maintains genetic diversity/Explores new search space regions
Crossover: Combines parent solutions/Creates new offspring/Propagates good characteristics.

Q2. Probability and Naive Bayes

[Marks 10 = 5+5, 30 min]

Part A. Consider the following Bayesian network. A, B, C, and D are Boolean random variables. If we know that A is true, what is the probability of D being true?

**Solution:**

$$\begin{aligned}
 P(D | A) &= \frac{P(A, D)}{P(A)} = \frac{P(A, B, C, D) + P(A, B, \neg C, D) + P(A, \neg B, C, D) + P(A, \neg B, \neg C, D)}{P(A)} \\
 &= P(B | A) P(C | A) P(D | B, C) + \\
 &\quad P(B | A) P(\neg C | A) P(D | B, \neg C) + \\
 &\quad P(\neg B | A) P(C | A) P(D | \neg B, C) + \\
 &\quad P(\neg B | A) P(\neg C | A) P(D | \neg B, \neg C) \\
 &= (0.2 \times 0.7 \times 0.3) + (0.2 \times 0.3 \times 0.25) + (0.8 \times 0.7 \times 0.1) + (0.8 \times 0.3 \times 0.35) \\
 &= 0.042 + 0.015 + 0.056 + 0.084 \\
 &= \mathbf{0.197} \text{ Answer}
 \end{aligned}$$

Another alternate way could be to follow the evidence. Once there is evidence on A, B and C are now independent. Find updated $P(B)$ and $P(C)$ with $P(A) = 1$ and $P(\neg A) = 0$. Now simply use $P(B)$ and $P(C)$ in calculation of $P(D)$ which is the sum of products of all the ways in which D can be achieved.

$$P(B) = (P(B|A) \times P(A)) + (P(B|\neg A) \times P(\neg A))$$

$$P(A) = 1 \text{ and } P(\neg A) = 0$$

$$P(B) = (P(B|A) \times P(A)) = 0.2 \Rightarrow P(\neg B) = 0.8$$

Similarly

$$P(C) = 0.7 \Rightarrow P(\neg C) = 0.3$$

$$P(D) = P(D|B, C)P(B)P(C) + P(D|\neg B, C)P(\neg B)P(C) + P(D|B, \neg C)P(B)P(\neg C) + P(D|\neg B, \neg C)P(\neg B)P(\neg C)$$

$$P(D) = 0.3 \times 0.2 \times 0.7 + 0.1 \times 0.8 \times 0.7 + 0.25 \times 0.2 \times 0.3 + 0.35 \times 0.8 \times 0.3$$

$$P(D) = 0.042 + 0.056 + 0.015 + 0.084 = 0.197$$

A financial institution is developing a system to predict whether a person will default on a loan based on the following categorical attributes:

Training data

Employed	Yes	Yes	No	Yes	No	No	Yes	Yes	No	No
Credit History	Good	Good	Bad	Bad	Good	Bad	Bad	Good	Good	Bad
Owns House	Yes	No	No	Yes	No	No	No	Yes	Yes	Yes
Default	No	No	Yes	No	Yes	Yes	No	No	No	Yes

Part B. Apply Naïve Bayes classifier to predict whether a new applicant with the following profile will default .

- Employed: **No**

- Credit History: **Bad**

- Owns House: **No**

Solution:

Step 1: Prior Probabilities for Target

Let $N = 10$ $P(\text{Yes}) = 4/10$, $P(\text{No}) = 6/10$

For Default

Employment:

Employment	Yes	No
yes	0/4	5/6
No	4/4	1/6

Credit History:

Credit History	Yes	No
Good	1/4	4/6
Bad	3/4	2/6

Own House:

Own House	Yes	No
yes	1/4	4/6
No	3/4	2/6

Classify Given Instance

Given:

- Employment = Unemployed
- Credit History = Bad
- Own House = No

For Default = Yes:

$$P(D = \text{yes}) \times P(\text{Unemployed} \mid D = \text{yes}) \times P(\text{Bad} \mid D = \text{yes}) \times P(\text{No} \mid D = \text{yes}) \\ = 0.4 \times (4/4) \times (3/4) \times (3/4) = 0.225$$

For Default = No:

$$P(D = \text{no}) \times P(\text{Unemployed} \mid D = \text{no}) \times P(\text{Bad} \mid D = \text{no}) \times P(\text{No} \mid D = \text{no}) \\ = 0.6 \times (1/6) \times (2/6) \times (2/6) = 0.01$$

Conclusion: $0.28 > 0.01 \rightarrow \text{Classify as Default} = \text{YES}$

Note:

No Laplace correction needed in this example. It is only required in cases where probabilities are zero.

For example: $P(\text{Employed} \mid \text{Default} = \text{yes}) = (0 + 1) / (4 + 2) = 1/6$

Q3. Supervised Learning**[10 marks, 5+5 , 45 min]**

Part A. A mobile food delivery startup wants to predict its monthly number of customer orders based on the amount of money spent on social media advertising. The company has collected data over the past 12 months on ad expenditure and the number of customer orders as shown below:

Month	1	2	3	4	5	6	7	8	9	10	11	12
Ad Expenditure	200	400	600	800	100	700	500	900	300	1000	200	600
Customer Orders	4	7	9	11	3	10	8	12	5	14	4	9

1. Calculate the linear regression equation and interpret the coefficients.
2. Estimate the average number of customer orders for a month where \$650 is spent on advertising. Using linear regression model.

Solution:

Convert Ad Expenditure to X values:

X (in \$100s)	2	4	6	8	1	7	5	9	3	10	2	6
Y (Orders)	4	7	9	11	3	10	8	12	5	14	4	9

Step 2: Compute Required Sums

$$\sum X=63 \quad \sum Y=96 \quad \sum XY=607 \quad \sum X^2=399 \quad n=12$$

Step 3: Find Regression Coefficients**Slope (a): $a \approx 1.51$ and Intercept (b): $B \approx 0.0725$** Final Linear Regression Equation: **$Y=1.51X+0.07$** **Interpretation of Coefficients:**

- **Slope (1.51):** For every additional \$100 spent on advertising, the number of customer orders is expected to increase by approximately 151.
- **Intercept (0.07):** If the company spends \$650 on ads, it can expect around 7 orders per month (theoretical starting point).

Part 2: Estimate Customer Orders for \$650 Ad Spend

$$X=650/100=6.5$$

$$Y=1.51(6.5)+0.07=9.815+0.07=9.89$$

The model predicts approximately **9.89 orders (i.e., ~989 orders)** for a \$650 ad spend.**OR**

Ad Expenditure (X)	200	400	600	800	100	700	500	900	300	1000	200	600
Customer Orders (Y)	4	7	9	11	3	10	8	12	5	14	4	9

Step1: Calculate Required Sums

$$n=12 \quad \sum X=6300 \quad \sum Y=96 \quad \sum XY=68000 \quad \sum X^2 = 4660000$$

Step2: Apply Linear Regression Formulas

$$M \approx 0.01302$$

$$B \approx 1.1645$$

Part 2: Prediction for \$650 ad spend:

$$Y=0.01302(650)+1.1645=8.463+1.1645=9.6275$$

So, the model predicts about 9.63 orders for a \$650 ad spend.

Part B. You are given the following training dataset with two features: **Exam Score** and **Project Score**.

The task is to classify a new student into either class "**Pass**" or "**Fail**" using the K-Nearest Neighbors (**KNN**) algorithm.

Student	Exam Score	Project Score	Class
A	85	70	Pass
B	80	90	Pass
C	60	60	Fail
D	70	65	Fail
E	95	85	Pass
F	55	50	Fail
G	75	70	Pass
H	50	45	Fail

A new student has **Exam Score = 72** and **Project Score = 68**.

- For **K = 3** and **K = 5**, determine the predicted class, after finding the Euclidean distance of the new student from all the training samples.
- Explain the effect of changing the value of K on the classification result.

Solution:

1. Euclidean Distances:

Distance to D: 3.61

Distance to G: 3.61

Distance to A: 13.15

Distance to C: 14.42

Distance to B: 23.41

Distance to F: 24.76

Distance to E: 28.6

Distance to H: 31.83

KNN Predictions:

For K = 3, the classes of nearest neighbors: ['Fail', 'Pass', 'Pass'] → Predicted class = Pass

For K = 5, the classes of nearest neighbors: ['Fail', 'Pass', 'Pass', 'Fail', 'Pass'] → Predicted class = Pass

2. Effect of Changing K:

When K is small (e.g., K=3), the algorithm is more sensitive to local noise in the data.

A single outlier may change the classification.

When K is larger (e.g., K=5), the classification is more stable and generalized.

but may overlook local patterns and boundaries.

CLO4: Summarize concepts and approaches in knowledge representation, planning, learning, robotics and other AI areas.

Q4. Supervised Learning

[10 marks, 5+5, 45 min]

Part A. You are given the following dataset of job applicants. The goal is to decide whether an applicant should be hired or not, using the ID3 decision tree algorithm.

Education	BS	BS	MS	PhD	PhD	BS	MS	PhD	MS	BS	MS	BS	PhD	BS
Experience	Low	Low	Med	High	High	Med	High	Med	Low	Med	Med	Low	High	Med
Skills	Poor	Good	Good	Best	Best	Poor	Good	Good	Good	Good	Best	Poor	Good	Best
Hired	No	No	Yes	Yes	Yes	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes

- Calculate the Entropy of the full dataset for the target attribute Hired.
- For each of the following attributes calculate the information gain:
 - Education
 - Experience
 - Skills
- Select the best attribute as the root node of the decision tree using the ID3 algorithm.

Solution Part A:**Step (a): Entropy of the Full Dataset**

Count of Hired: - Yes = 9 - No = 5 Total = 14

$$\begin{aligned}\text{Entropy}(S) &= -p(\text{Yes}) * \log_2(p(\text{Yes})) - p(\text{No}) * \log_2(p(\text{No})) \\ &= - (9/14) * \log_2(9/14) - (5/14) * \log_2(5/14) \\ &\approx -0.643 * -0.643 - 0.357 * -1.485 = 0.940\end{aligned}$$

Step (b): Information Gain for Each Attribute**1. Education**

Bachelors: 6 → 2 Yes, 4 No → Entropy = 0.918 Masters: 4 → 3 Yes, 1 No → Entropy = 0.811.

PhD: 4 → 4 Yes → Entropy = 0

$$\begin{aligned}\text{Gain}(\text{Education}) &= 0.940 - [6/14 * 0.918 + 4/14 * 0.811 + 4/14 * 0] \\ &= 0.940 - (0.626) = 0.313\end{aligned}$$

2. Experience

Low: 4 → 0 Yes, 4 No → Entropy = 0 Medium: 5 → 4 Yes, 1 No → Entropy = 0.722

High: 5 → 5 Yes → Entropy = 0

$$\begin{aligned}\text{Gain}(\text{Experience}) &= 0.940 - [4/14 * 0 + 5/14 * 0.722 + 5/14 * 0] \\ &= 0.940 - 0.258 = 0.682\end{aligned}$$

3. Skills

Poor: 3 → 0 Yes, 3 No → Entropy = 0 Good: 7 → 5 Yes, 2 No → Entropy ≈ 0.863

Excellent: 4 → 4 Yes → Entropy = 0

$$\begin{aligned}\text{Gain}(\text{Skills}) &= 0.940 - [3/14 * 0 + 7/14 * 0.863 + 4/14 * 0] \\ &= 0.940 - 0.431 = 0.509\end{aligned}$$

Step (c): Best Attribute for Root Node

Information Gain Summary:

- Education: 0.289 - Experience: 0.682 → Selected - Skills: 0.509
→ Best attribute = Experience

Part B. You are working as a data scientist in a healthcare startup that has developed a machine learning model to predict whether a patient has a rare disease (Positive) or not (Negative) based on their medical records. The disease is quite rare, affecting only about 2% of the population.

The model was tested on a dataset of 1,000 patients, and it made the following predictions:

- It predicted 950 patients as Negative, and out of these, 940 were actually healthy, but 10 were sick.
- It predicted 50 patients as Positive, and out of these, 30 were healthy, but 20 were actually sick.

Perform the following:

1. Construct the Confusion Matrix based on the information provided.
2. Calculate the accuracy of the model.

Solution Part B:

	Predicted Positive	Predicted Negative
Actual Positive	20 (TP)	10 (FN)
Actual Negative	30 (FP)	940 (TN)

$$\text{Accuracy} = (TP + TN) / \text{Total} = (20 + 940) / 1000 = 0.96 \text{ or } 96\%$$

Q5. K-Means and ANN**[Marks 10, 5+5 25 min]**

Part A. In a seismic monitoring system, each sensor node records its x, y location in kilometers. Your task is to cluster sensor nodes into TWO zones using the K-Means algorithm to optimize alert response regions.

You are given the following 8 sensor coordinates:

Sensor	1	2	3	4	5	6	7	8
X(km)	5	6	40	38	8	47	43	7
Y(km)	15	17	60	58	14	66	62	13

Simulate the K-Means algorithm for $k = 2$. Perform one iteration with data point 1 and 6 as initial cluster centers, and explain:

- Which sensors belong to which cluster.

- What will be the updated cluster centers after one iteration.

Solution:

Initial Cluster Centres:

Cluster 1 Center (C1): Sensor #1 $\rightarrow (5, 15)$

Cluster 2 Center (C2): Sensor #6 $\rightarrow (47, 66)$

Step A: One Iteration Assignment

We'll compute distances from each point to C1 and C2 using Euclidean distance:

Step 1: Initial Cluster Centres

Cluster 1 (C1) is initialised at Sensor 1: (5, 15)

Cluster 2 (C2) is initialised at Sensor 6: (47, 66)

Step 2: Assign each sensor to the nearest cluster

We calculate the Euclidean distance from each sensor to the two cluster centres:

Sensor 1 (5, 15):

Distance to C1 = $\sqrt{((5-5)^2 + (15-15)^2)} = 0 \rightarrow$ Assign to Cluster 1

Sensor 2 (6, 17):

Distance to C1 = $\sqrt{((6-5)^2 + (17-15)^2)} = \sqrt{1^2 + 2^2} = \sqrt{5} \approx 2.24$

Distance to C2 = $\sqrt{((6-47)^2 + (17-66)^2)} = \sqrt{(-41)^2 + (-49)^2} = \sqrt{1681 + 2401} \approx \sqrt{4082} \approx 63.9 \rightarrow$ Assign to Cluster 1

Sensor 3 (40, 60):

Distance to C1 = $\sqrt{((40-5)^2 + (60-15)^2)} = \sqrt{1225 + 2025} = \sqrt{3250} \approx 57$

Distance to C2 = $\sqrt{((40-47)^2 + (60-66)^2)} = \sqrt{49 + 36} = \sqrt{85} \approx 9.22 \rightarrow$ Assign to Cluster 2

Sensor 4 (38, 58):

Distance to C1 = $\sqrt{((38-5)^2 + (58-15)^2)} = \sqrt{1089 + 1849} = \sqrt{2938} \approx 54.2$

Distance to C2 = $\sqrt{((38-47)^2 + (58-66)^2)} = \sqrt{81 + 64} = \sqrt{145} \approx 12.04 \rightarrow$ Assign to Cluster 2

Sensor 5 (8, 14):

Distance to C1 = $\sqrt{((8-5)^2 + (14-15)^2)} = \sqrt{9 + 1} = \sqrt{10} \approx 3.16$

Distance to C2 = $\sqrt{((8-47)^2 + (14-66)^2)} = \sqrt{1521 + 2704} = \sqrt{4225} = 65 \rightarrow$ Assign to Cluster 1

Sensor 6 (47, 66):

Distance to C1 = $\sqrt{((47-5)^2 + (66-15)^2)} = \sqrt{1764 + 2601} = \sqrt{4365} \approx 66.1$

Distance to C2 = 0 (it is the center) $\rightarrow$ Assign to Cluster 2

Sensor 7 (43, 62):

Distance to C1 = $\sqrt{((43-5)^2 + (62-15)^2)} = \sqrt{1444 + 2209} = \sqrt{3653} \approx 60.4$

Distance to C2 = $\sqrt{((43-47)^2 + (62-66)^2)} = \sqrt{16 + 16} = \sqrt{32} \approx 5.66 \rightarrow$ Assign to Cluster 2

Sensor 8 (7, 13):

Distance to C1 = $\sqrt{((7-5)^2 + (13-15)^2)} = \sqrt{4 + 4} = \sqrt{8} \approx 2.83$

Distance to C2 = $\sqrt{((7-47)^2 + (13-66)^2)} = \sqrt{1600 + 2809} = \sqrt{4409} \approx 66.4 \rightarrow$ Assign to Cluster 1

Cluster Assignments After First Iteration

Cluster 1: Sensors 1, 2, 5, 8

Cluster 2: Sensors 3, 4, 6, 7

Step 3: Recalculate New Cluster Center

Cluster 1: Sensor coordinates: (5,15), (6,17), (8,14), (7,13)

Average X = $(5 + 6 + 8 + 7) / 4 = 26 / 4 = 6.5$ Average Y = $(15 + 17 + 14 + 13) / 4 = 59 / 4 = 14.75$

→ New Cluster 1 Center = (6.5, 14.75)

Cluster 2: Sensor coordinates: (40,60), (38,58), (47,66), (43,62)

Average X = $(40 + 38 + 47 + 43) / 4 = 168 / 4 = 42.0$ Average Y = $(60 + 58 + 66 + 62) / 4 = 246 / 4 = 61.5$

→ New Cluster 2 Center = (42.0, 61.5)

Final Result (After One Iteration)

Cluster 1 Sensors: 1, 2, 5, 8 → New center: (6.5, 14.75)

Cluster 2 Sensors: 3, 4, 6, 7 → New center: (42.0, 61.5)

Part B. A logistics company wants to use an artificial neural network (ANN) to predict the optimal delivery category for a package based on certain attributes provided at the time of dispatch. Each package is described using the following features:

- Package weight (in kilograms) - Dimensions (in cubic centimeters)
- Delivery distance (in kilometers) - Package fragility score (0 to 10)
- Delivery speed preference (standard, express, overnight — encoded numerically)

The possible delivery categories are

- Standard Ground - Express Air - Overnight Priority - Same-Day Urban

The company has several thousand historical delivery records and wants to automate this process.

Answer the following questions:

- How many input values (units) will the ANN need?
Input Layer (5 units): Package weight, dimensions, delivery distance, package fragility score, and delivery speed preference.
- How many output values (units) will the ANN need?
Output Layer (4 units): Standard Ground, Express Air, Overnight Priority, Same-Day Urban
- Should the ANN use hidden layers or not? Why?
Yes, hidden units should be used. Reason: The mapping from input features to delivery categories is likely nonlinear (e.g., the effect of distance and fragility may interact). Hidden layers allow the model to learn complex decision boundaries.
- What activation function is commonly used in the hidden layer?
ReLU — it's simple and works well in practice.
- Draw a simple feedforward neural network diagram for the above process.

